**55.** Let $G$ be cyclic and let $a$ be a generator for $G$. For $x, y \in G$, there exist $m, n \in \mathbb{Z}$ such that $x = a^m$ and $y = b^n$. Then $xy = a^m b^n = a^{m+n} = a^{n+m} = a^n a^m = yx$, so $G$ is abelian.

**56.** We can show it if $G$ is abelian. Let $a, b \in G$ so that $a^n, b^n \in G_n$. Then $a^n b^n = (ab)^n$ because $G$ is abelian, so $G_n$ is closed under the induced operation. Also $e = e^n \in G_n$. Finally $(a^n)^{-1} = (a^{-1})^n \in G_n$, so $G_n$ is indeed a subgroup of $G$.

**57.** Let $G$ be a group with no proper nontrivial subgroups. If $G = \{e\}$, then $G$ is of course cyclic. If $G \neq \{e\}$, then let $a \in G, a \neq e$. We know that $\langle a \rangle$ is a subgroup of $G$ and $\langle a \rangle \neq \{e\}$. Because $G$ has no proper nontrivial subgroups, we must have $\langle a \rangle = G$, so $G$ is indeed cyclic.

## 6. Cyclic Groups

**1.** $42 = 9 \cdot 4 + 6, q = 4, r = 6$ **2.** $-42 = 9(-5) + 3, q = -5, r = 3$ **3.** $-50 = 8(-7) + 6, q = -7, r = 6$

**4.** $50 = 8 \cdot 6 + 2, q = 6, r = 2$ **5.** 8 **6.** 8 7. 60

**8.** 1, 2, 3, and 4 are relative prime to 5 so the answer is 4.

**9.** 1, 3, 5, and 7 are relatively prime to 8 so the answer is 4.

**10.** 1, 5, 7, and 11 are relatively prime to 12 so the answer is 4.

**11.** 1, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 49, 53,and 59 are relatively prime to 60 so the answer is 16.

**12.** There is one automorphism; 1 must be carried into the only generator which is 1.

**13.** There are 2 automorphisms; 1 can be carried into either of the generators 1 or 5

**14.** There are 4 automorphisms; 1 can be carried into any of the generators 1, 3, 5, or 7.

**15.** There are 2 automorphisms; 1 can be carried into either of the generators 1 or -1.

**16.** There are 4 automorphisms; 1 can be carried into any of the generators 1, 5, 7, or 11.

**17.** gcd(25, 30) = 5 and 30/5 = 6 so $\langle 25 \rangle$ has 6 elements.

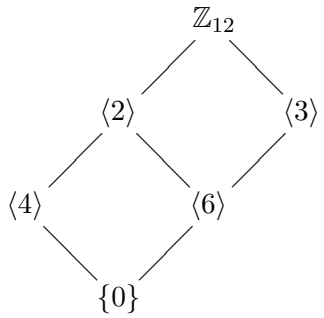
**18.** gcd(30, 42) = 6 and 42/6 = 7 so $\langle 30 \rangle$ has 7 elements.

**19.** The polar angle for $i$ is $\pi/2$, so it generates a subgroup of 4 elements.

**20.** The polar angle for $(1 + i)/\sqrt{2}$ is $\pi/4$, so it generates a subgroup of 8 elements.

**21.** The absolute value of $1 + i$ is $\sqrt{2}$, so it generates an infinite subgroup of $\aleph_0$ elements.

**22.** Subgroup diagram:




23. (See the answer in the text.)

**24.** Subgroup diagram:




**25.** 1, 2, 3, 6 **26.** 1, 2, 4, 8 **27.** 1, 2, 3, 4, 6, 12 **28.** 1, 2, 4, 5, 10, 20 **29.** 1, 17

**30.** Incorrect; $n$ must be minimal in $\mathbb{Z}^+$ with that property.

An element $a$ of a group $G$ has **order** $n \in \mathbb{Z}^+$ if $a^n = e$ and $a^m \neq e$ for $m \in \mathbb{Z}^+$ where $m < n$.

**31.** The definition is correct.

**32.** T F F F T F F F T T f) The Klein 4-group is an example. g) 9 generates $\mathbb{Z}_{20}$.

**33.** The Klein 4-group **34.** $\langle \mathbb{R}, + \rangle$ **35.** $\mathbb{Z}_2$

**36.** No such example exists. Every infinite cyclic group is isomorphic to $\langle \mathbb{Z}, + \rangle$ which has just two generators, 1 and -1.

**37.** $\mathbb{Z}_8$ has generators 1, 3, 5, and 7. **38.** $i$ and $-i$

**39.** Corresponding to polar angles $n(2\pi/6)$ for $n = 1$ and 5, we have $\frac{1}{2}(1 \pm i\sqrt{3})$.

**40.** Corresponding to polar angles $n(2\pi/8)$ for $n =$ 1,7,3, and 5, we have $\frac{1}{\sqrt{2}}(1 \pm i)$ and $\frac{1}{\sqrt{2}}(-1 \pm i)$.

**41.** Corresponding to polar angles $n(2\pi/12)$ for $n =$ 1,11,5, and 7, we have $\frac{1}{2}(\sqrt{3} \pm i)$ and $\frac{1}{2}(-\sqrt{3} \pm i)$.

**42.** Expressing two elements of the group as powers of the same generator, their product is the generator raised to the sum of the powers, and addition of integers is commutative.

**43.** Asuming the subgroup isn't just $\{e\}$, let $a$ be a generator of the cyclic group, and let $n$ be the smallest positive integer power of $a$ that is in the subgroup. For $a^m$ in the subgroup, use the division algorithm for $n$ divided by $m$ and the choice of $n$ to argue that $n = qm$ for some integer $q$, so that $a^m = (a^n)^q$.

**44.** By the homomorphism property $\phi(ab) = \phi(a)\phi(b)$ extended by induction, we have $\phi(a^n) = (\phi(a))^n$ for all $n \in \mathbb{Z}+$. By Theorem 3.14, we know that $\phi(a^0) = \phi(e) = e'$. The equation $e' = \phi(e) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1})$ shows that $\phi(a^{-1}) = (\phi(a))^{-1}$. Extending this last equation by induction, we see that $\phi(a^{-n}) = (\phi(a))^{-n}$ for all negative integers $-n$. Because $G$ is cyclic with generator $a$, this means that for all $g = a^n \in G, \phi(g) = \phi(a^n) = [\phi(a)]^n$ is completely determined by the value $\phi(a)$.

**45.** The equation $(n_1r + m_1s) + (n_2r + m_2s) = (n_1 + n_2)r + (m_1 + m_2)s$ shows that the set is closed under addition. Because $0r + 0s = 0$, we see that 0 is in the set. Because $[(-m)r + (-n)s] + (mr + ns) = 0$, we see that the set contains the inverse of each element. Thus it is a subgroup of $\mathbb{Z}$.

**46.** Let $n$ be the order of $ab$ so that $(ab)^n = e$. Multiplying this equation on the left by $b$ and on the right by $a$, we find that $(ba)^{n+1} = bea = (ba)e$. Cancellation of the first factor $ba$ from both sides shows that $(ba)^n = e$, so the order of $ba$ is $\leq n$. If the order of $ba$ were less than $n$, a symmetric argument would show that the order of $ab$ is less than $n$, contrary to our choice of $n$. Thus $ba$ has order $n$ also.

**47. a.** As a subgroup of the cyclic group $\langle \mathbb{Z}, + \rangle$, the subgroup $G = r\mathbb{Z} \cap s\mathbb{Z}$ is cyclic. The positive generator of $G$ is the **least common multiple** of $r$ and $s$.

**b.** The least common multiple of $r$ and $s$ is $rs$ if and only if $r$ and $s$ are relative prime, so that they have no common prime factor.

**c.** Let $d = ir + js$ be the gcd of $r$ and $s$, and let $m = kr = qs$ be the least common multiple of $r$ and $s$. Then $md = mir + mjs = qsir + krjs = (qi + kj)rs$, so $rs$ is a divisor of $md$. Now let $r = ud$ and let $s = vd$. Then $rs = uvdd = (uvd)d$, and $uvd = rv = su$ is a multiple of $r$ and $s$, and hence $uvd = mt$. Thus $rs = mtd = (md)t$, so $md$ is divisor of $rs$. Hence $md = rs$.

**48.** Note that every group is the union of its cyclic subgroups, because every element of the group generates a cyclic subgroup that contains the element. Let $G$ have only a finite number of subgroups, and hence only a finite number of cyclic subgroups. Now none of these cyclic subgroups can be infinite, for every infinite cyclic group is isomorphic to $\mathbb{Z}$ which has an infinite number of subgroups, namely $\mathbb{Z}, 2\mathbb{Z}, 3\mathbb{Z}, \cdots$. Such subgroups of an infinite cyclic subgroup of $G$ would of course give an infinite number of subgroups of $G$, contrary to hypothesis. Thus $G$ has only finite cyclic subgroups, and only a finite number of those. We see that the set $G$ can be written as a finite union of finite sets, so $G$ is itself a finite set.

**49.** The Klein 4-group $V$ is a counterexample.

**50.** Note that $xax^{-1} \neq e$ because $xax^{-1} = e$ would imply that $xa = x$ and $a = e$, and we are given that $a$ has order 2. We have $(xax^{-1})^2 = xax^{-1}xax^{-1} = xex^{-1} = xx^{-1} = e$. Because $a$ is given to be the *unique* element in $G$ of order 2, we see that $xax^{-1} = a$, and upon multiplication on the right by $x$, we obtain $xa = ax$ for all $x \in G$.

**51.** The positive integers less that $pq$ and relatively prime to $pq$ are those that are not multiples of $p$ and are not multiples of $q$. There are $p - 1$ multiples of $q$ and $q - 1$ multiples of $p$ that are less than $pq$. Thus there are $(pq - 1) - (p - 1) - (q - 1) = pq - p - q + 1 = (p - 1)(q - 1)$ positive integers less than $pq$ and relatively prime to $pq$.